Options Strategies

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Abstract

1 Box Spread

> "A box spread is a combination of a bull call spread with strike prices K_1 and K_2 and a bear put spread with the same two strike prices."
>
> – John C. Hull

A box spread can be decomposed into four options:

- Long call at strike K_1;
- Short call at strike K_2;
- Long put at strike K_2;
- Short put at strike K_1.

where $K_1 < K_2$, which is equivalent to combining a bull call spread with a bear put spread over the same strikes and expiry.

2 Data

All the data will be retrieved from Workspace (Refinitiv-Reuters), Bloomberg and via yfinance .

2.1 Data Description

The chosen ticker is VIX ($.VIX), formally:

> CBOE VIX Index is a real-time market index that represents the market's expectation of 30-day forward-looking volatility. Derived from the price inputs of the S&P 500 index options, it provides a measure of market risk and investors' sentiments.
>
> — Refinitiv, Reuters.

Since the listed options from the ticker above are European style (see[1]) and we are pricing them via Black and Scholes (1973), it is intuitive to use an appropriate ticker that the market prices already reflect most of the BSM model assumptions.

We can think of other reasons to use VIX as a fairly good candidate for this case, mainly high liquidity (see Appendix A)) across tenors and by being a underlying that pays no dividends (see Appendix B) as we're not interested on the overly complicated resolution to pricing options with cash dividends.

The VIX is currently a very useful benchmark on the expected 30-day volatility, we can also calculate the VIX with other maturities, see below:

Figure 1: VIX - VIX3M - VIX6M

Here for example we can see some term structure inversion:[2]

Figure 2: Spread VIX - VIX6M

The VIX[3] also tends to overestimate subsequently realized volatility, as shown below. Part of the

[1] https://www.cboe.com/tradable-products/vix/vix-options/specifications

[2] When short term IV is above long term IV.

[3] Cboe Volatility Index Mathematics Methodology, available at https://cdn.cboe.com/resources/indices/Cboe_Volatility_Index_Mathematics_Methodology.pdf.

gap is due to the index computation as the square root of a strike-weighted portfolio of OTM options replicating the payoff of a variance swap, so the convexity contribution of the wings places it above ATM implied volatility. The remaining, persistent component reflects the variance risk premium (Carr and Wu, 2009).

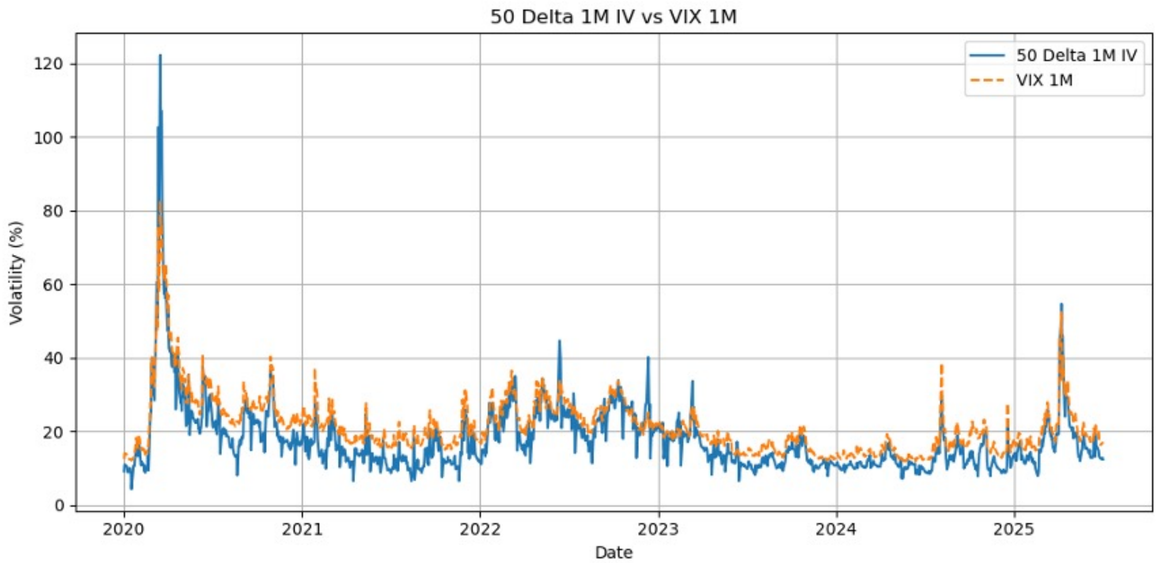

Figure 3: SPX 50 Delta 1M vs VIX

3 BSM and Black's 76 Model

Let a general market model $\mathcal{M}$ in discrete time with: a finite state space Ω, a filtration $\mathcal{F}$, a strictly positive probability measure P defined on $\wp(\Omega)$, a terminal date $T \in \mathbb{N}$, $T < \infty$ and a set $\mathbb{S} \equiv \{(S_t^k)_{t \in \{0,\ldots,T\}} : k \in \{0, \ldots, K\}\}$ of $K+1$ strictly positive security price processes. Together one has $\mathcal{M} = \{(\Omega, \wp(\Omega), \mathbb{F}, P), T, \mathbb{S}\}$.

The BSM model follows a Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t \tag{1}$$

where W_t is a standard Brownian motion.

We have:

$$C = S_0 \, N(d_1) - K e^{-rT} N(d_2) \tag{2}$$

where:

$$d_1 = \frac{\ln\left(\dfrac{S_0}{K}\right) + \left(r + \dfrac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} \tag{3}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{4}$$

The original model has some particular assumptions; most of them although not as realistic as real market behaviors but surprisingly converging to real prices. The only *modification* we need here is to change our measurable spot price into futures prices as VIX is not directly traded. It's clear by now that Black (1976) is the correct model, so essentially rewriting equation 2 we have:

$$C = e^{-rT}\left[F\,N(d_1) - K\,N(d_2)\right] \tag{5}$$

where:

$$d_1 = \frac{\ln\left(\frac{F}{K}\right) + \frac{\sigma^2}{2}T}{\sigma\sqrt{T}} \tag{6}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{7}$$

4 Pricing with Black's Model

With a maturity of 28 days[4], $\sigma = 100\%$[5], $F = 20$, $r = 4\%$, we have two subsets of strikes: $K_1 = 15, K_2 = 25$.

Using `Quantlib`:

Position	**Strike**	**Price**
Long call	K_1	5.3524
Short call	K_2	0.7307
Short put	K_1	0.3677
Long put	K_2	5.7153

Table 1: Box Spread Option Positions and Prices

Our box spread value should be equal to a call spread plus a put spread, resulting in a box value of 9.9694. Some sensitivities for our box:

Greek	**Value**
Forward Delta	0.000000
Gamma	0.000000
Vega	0.000000
Rho	−0.764773
Theta	0.398774
Theta per Day	0.001093
Probability ITM	0.643443

Table 2: Box Spread Greeks

One can also input a sort of historical volatility model taking for example the EWMA (Exponentially Weighted Moving Average) from the VXN6 futures data:

$$\hat{\sigma}_t^2 = (1-\alpha)\sum_{i=0}^{\infty} \alpha^i\, r_{t-i}^2 \tag{8}$$

where r_t are log-returns. The parameter α is related to the span s by $\alpha = 1 - 2/(s+1)$. With 3 months of data and arranged in 10-minute bar OHLC for VXN6. A trading day has $s = 39$ bars:

[4] We'll remain with the Calendar days / 365 as it's explicit in the methodology.

[5] The implied volatility of VIX is proxied by the Cboe VVIX Index (`^VVIX`), which measures the expected volatility of the 30-day forward price of VIX, i.e. a volatility of volatility. Available at https://cdn.cboe.com/resources/indices/documents/vvix-termstructure.pdf.

$$\hat{\sigma}_{\text{EWMA}} = \hat{\sigma}_t \times \sqrt{252 \times 39} \tag{9}$$

Span	σ
39 bars $\approx$ 1 day	35.91%
195 bars $\approx$ 1 week	38.81%

Table 3: EWMA estimated volatility

With the same setup as before but updating the risk free rate as explained in chapter 5.2 and with a $\hat{\sigma}_{\text{EWMA}} = 35.91\%$ we have:

Position	**Strike**	**Price**
Long call	K_1	3.9957
Short call	K_2	0.0285
Short put	K_1	0.0076
Long put	K_2	4.0165

Table 4: Box Spread Option Positions and Prices (EWMA)

The above results in a box value of 7.9760[6]. Some sensitivities for our box:

Greek	**Value**
Forward Delta	0.000000
Gamma	0.000000
Vega	0.000000
Rho	−0.611818
Theta	0.319020
Theta per Day	0.000874
Probability ITM	0.534874

Table 5: Box Spread Greeks (EWMA)

5 Pricing with Market Data

5.1 Implied Volatility

Usually we backup the IV from quoted market prices, but there are several ways to estimate implied volatility, Brenner and Subrahmanyam (1988) provided a closed form estimate of IV, by using a initial estimate:

$$\sigma \approx \sqrt{\frac{2\pi}{T} \cdot \frac{C}{S}} \tag{10}$$

Although BSM provides a closed-form pricing formula, there is no closed-form inverse for it, but because it has a closed-form vega, and the derivative is nonnegative, we can use the Newton-Raphson formula with (some[7]) confidence. Starting from an initial guess σ_0, we iterate

[6] As we changed other parameters, mainly strikes with multiples of forward price and risk free rate it is obvious the box spread price changes. If this would remain the same, simply changing the vol would be indifferent as vega shows us.

[7] Newton-Raphson might fail to converge for deep ITM options.

$$\sigma_{n+1} = \sigma_n - \frac{BS(\sigma_n) - P}{\nu(\sigma_n)} \tag{11}$$

Until the solution reaches sufficient accuracy. This approach is valid for vanilla options where BSM yields a closed-form solution. For exotic payoffs or American-exercise options, vega may not be available in closed form, and more robust root-finding methods such as Brent's method are typically preferred.

For further reading on implied volatility computation, see Jäckel (2006) and Jäckel (2013) offer elegant treatments of the inversion problem, the latter introducing a fully rational approximation that avoids iterative methods altogether. Li and Lee (2009) propose an adaptive successive over-relaxation scheme that improves convergence robustness, while Stefanica and Radoicic (2017) derive a closed-form approximation that is both analytically tractable and computationally efficient.

Recently Schadner (2026) did an explicit formula for BSM IV were the key insight is that the normalized call price can be written as a survival probability of an inverse Gaussian distribution, so that inverting this identity yields implied volatility directly through the corresponding quantile function, requiring no initial guess, iteration, or approximation.

5.2 Risk Free Rate

It is no surprise by now that government bond instruments are not adequate for pricing derivatives. The work of Feldhütter and Lando (2008) explain the reasons behind why OIS rates are preferred. For e.g., Bloomberg derivatives tools (`OVME`, `OVML`, `SWPM`, `DLIB`,...) use overnight OIS instruments; taking an EUR funding account we can choose from: `EUR Depo, IRS vs 6M` for EURIBOR-linked instruments; `EUR ESTR Swap ZC` as the CSA-consistent OIS discount curve; `EUR EURIBOR Swap ZC` for projecting EURIBOR floating cashflows; and futures-based alternatives `EUR FEI Future ZC` (3M EURIBOR futures) and `EUR FEU3 Future ZC` (€STR futures).

The objective of this work is the rapid code modification and it's user interchangeability (e.g., tickers, strategies, tenors,...) without proprietary models or paid platforms e.g., Bloomberg, Reuters. So there is a certain preference on using `yfinance`. As the standard is SOFR OIS, and these are traded OTC, the proxy available here is SOFR CME Futures[8].

5.3 Results

Implied volatilities are recovered by numerically inverting the Black–76 pricing function. Since the model price is strictly monotonic in σ, each quote admits a unique implied vol within the no-arbitrage range we solve

$$f(\sigma) = \text{Black76}(\sigma) - P_{\text{mkt}} = 0 \tag{12}$$

with Brent's method, bracketed on $\sigma \in [0,\, 1000\%]$, returning `NaN` when a quote falls outside the no-arbitrage bounds.

Each contract's price is the bid–ask mid where the quote is two-sided and the last trade otherwise, retaining only liquid contracts (open interest ≥ 10 or volume ≥ 1). The forward is taken from put–call parity at the near-the-money strike:

$$F = K + e^{rT}\big(C - P\big), \tag{13}$$

We build the smile from out-of-the-money quotes only: puts for $K < F$, calls for $K > F$. Parity makes the two sides equivalent. We drop quotes with relative spread above 25% and keep strikes in $[0.6F,\, 3F]$.

As a check, we reprice a box from market mids against its arbitrage-free value,

$$V_{\text{box}} = (C_{K_1} - C_{K_2}) + (P_{K_2} - P_{K_1}) = e^{-rT}(K_2 - K_1). \tag{14}$$

[8]SOFR futures rates are acceptable for the very short end but require a convexity adjustment; however, given that rho is negligible for short-dated options, the choice has no material pricing impact.

The residual ($\approx +0.0190$ on an eight-point box) is within the bid–ask, so the cleaned quotes are arbitrage-free.

Quantity	**IV**
OTM IV, $K_1 = 16$ (put)	64.65%
OTM IV, $K_2 = 24$ (call)	112.43%
Call spread ($C_{K_1} - C_{K_2}$)	2.8150
Put spread ($P_{K_2} - P_{K_1}$)	5.1800
Box value (market)	**7.9950**
Theoretical $(K_2 - K_1)e^{-rT}$	7.9760
Arbitrage gap	+0.0190

Table 6: Market results for the VXN6 expiry (22 July 2026).

Greek	Value
Forward Delta	0.000000
Gamma	0.000000
Vega	0.000000
Rho	-0.611861
Theta	0.311863
Theta per Day	0.000854
Probability ITM	0.642841

Table 7: Box spread Greeks (IV)

6 Some Conclusions

The box behaves like a simple loan. No matter which volatility we put in the VVIX proxy (100%), the EWMA estimate (35.91%), or the values implied by market prices the box is always worth the same: $(K_2 - K_1)e^{-rT}$. Pricing the box from real quotes yields 7.9950, against a theoretical 7.9760 a gap of only 0.0190, small enough to sit inside the bid–ask spread. In other words.

All of this only works with European options. Box spreads only lock in a riskless payoff with European options. E.g., CBOE equity options are American, so early exercise risk breaks the arbitrage, selling the box for less than its true American-option value (bull call spread + bear put spread) is a losing trade, since the deep ITM short put gets exercised against by the other counterpart almost immediately.

A Commitments of Traders – CBOE Futures Exchange

VIX FUTURES – CBOE FUTURES EXCHANGE

Commitments of Traders with Delta-adjusted Options and Futures Combined, June 02, 2026

	Total Open Interest	Non-Commercial			Commercial		Total		Nonreportable	
		Long	Short	Spreading	Long	Short	Long	Short	Long	Short
All	410,627	111,052	181,004	84,123	189,934	117,261	385,109	382,388	25,518	28,239
Old	410,627	111,052	181,004	84,123	189,934	117,261	385,109	382,388	25,518	28,239
Other	0	0	0	0	0	0	0	0	0	0
Changes in Commitments from: May 26, 2026										
	26,058	−771	14,697	8,388	15,764	321	23,381	23,406	2,677	2,652
Percent of Open Interest Represented by Each Category of Trader										
All	100.0	27.0	44.1	20.5	46.3	28.6	93.8	93.1	6.2	6.9
Old	100.0	27.0	44.1	20.5	46.3	28.6	93.8	93.1	6.2	6.9
Other	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Number of Traders in Each Category										
All	184	59	65	80	43	42	150	146		
Old	184	59	65	80	43	42	150	146		
Other	0	0	0	0	0	0	0	0		

B Are Dividend Yields Irrelevant?

Even though it is clear by now the importance of a discrete formulation for dividends (see e.g., Whaley (1981), Longstaff and Schwartz (2001), Bos and Vandermark (2002), Frishling (2002), Klassen (2015), Guennoun and Henry-Labordère (2017), Healy (2021)), there are still some situations where a continuous yield q might be useful.

For example, obtaining an exact amount of future cash dividends can be complicated. Today that projection can be done via internal forecasts or other platforms, e.g., Bloomberg.[9] But even so, for long tenors a blended scheme is commonly used.[10] As a result, dividends are commonly modeled as discrete cash payments in the short term, a combination of discrete cash and proportional (i.e., stock-based) dividends in the medium term, and as a continuous dividend yield in the long term.[11]

[9]The preference for using Bloomberg's BDVD dividend forecasts over statistical models is justified by its robust methodology, which combines seven key factors: (1) company guidance, (2) dividend health score, (3) industry analysis, (4) historical trend analysis, (5) market analysts' consensus estimates, (6) options market implied dividends, and (7) historical regression analysis of fundamentals compared to dividends.

[10]For forward-starting options on dividends one cannot use a static blending scheme conditional on future spot, as dividend volatility would be too large.

[11]It is obvious by now that a dividend forecast risk premium for special dividends is ignored.

Figure 4: Bloomberg EQD Dividend Settings

There's still some pricing models for vanilla options with cash, mainly the spot adjustment, strike adjustment, spot-strike adjustment altough most OMMs and EQD desks use a sort of blending scheme as shown above.

References

Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2):167–179.

Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *The Journal of Political Economy*, 81(3):637–654.

Bos, M. and Vandermark, S. (2002). Modeling european options with discrete dividends. Technical report, ABN AMRO Quantitative Research.

Brenner, M. and Subrahmanyam, M. G. (1988). A simple formula to compute the implied standard deviation. *Financial Analysts Journal*, 44(5):80–83.

Carr, P. and Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3):1311–1341.

Feldhütter, P. and Lando, D. (2008). Decomposing swap spreads. *Journal of Financial Economics*, 88(2):375–405.

Frishling, V. (2002). A discrete question. *Risk Magazine*, 15:115–116.

Guennoun, H. and Henry-Labordère, P. (2017). Equity modeling with stochastic dividends. SSRN Working Paper No. 2960141.

Healy, J. (2021). The pricing of vanilla options with cash dividends as a classic vanilla basket option problem. arXiv preprint arXiv:2106.12971.

Jäckel, P. (2006). By implication.

Jäckel, P. (2013). Let's be rational.

Klassen, T. R. (2015). Pricing vanilla options with cash dividends. Unpublished manuscript.

Li, M. and Lee, K. (2009). An adaptive successive over-relaxation method for computing the Black–Scholes implied volatility.

Longstaff, F. A. and Schwartz, E. S. (2001). Valuing american options by simulation: A simple least-squares approach. *Review of Financial Studies*, 14(1):113–147.

Schadner, W. (2026). An explicit solution to Black–Scholes implied volatility. arXiv:2604.24480v1 [q-fin.MF].

Stefanica, D. and Radoicic, R. (2017). An explicit implied volatility formula. *International Journal of Theoretical and Applied Finance*.

Whaley, R. (1981). On the valuation of american call options on stocks with known dividends. *Journal of Financial Economics*, 9:207–211.